

ROUND 3 ANALYTICAL REPORT

Data Vortex A'26 — Round 3 | Team: hy4366

Assigned topic: Public Reaction to Changes in the Recommendation Algorithm

Executive Summary

This reproducible analysis uses 204 genuine, retained public Hacker News archive records. Predictions are from the preserved Round 2 TF-IDF + Logistic Regression pipeline, not hand labels. Results describe this technical-community source only.

DATA COLLECTION METHOD

Source: Hacker News public Algolia archive API. Endpoint: <https://hn.algolia.com/api/v1/search>. Queries: Twitter recommendation algorithm, YouTube recommendation algorithm, Instagram algorithm change, feed algorithm, recommendation algorithm, recommendation ranking, Facebook news feed algorithm, algorithmic feed. The existing archive export was not recollected during this correction. A second documented screen retained records with recommendation/feed context plus a concrete change, reaction, quality, or user-control signal.

TIME WINDOW

Collection ran from 2026-09-21T19:44:34.808161+00:00 to 2026-09-21T19:44:49.802497+00:00 (UTC). Retained source observation window: 2011-02-02 06:34:14+00:00 to 2026-09-21 18:49:57+00:00 UTC. This is a historical public archive sample collected through the Hacker News Algolia archive API; it is not a continuous live stream.

Dataset Overview and Data Quality

Initial screened records: 389. Excluded as not specific to change/reaction: 182; duplicate cleaned text excluded: 5; final retained records: 204. Score/comment fields are unavailable for most HN comments and remain missing.

Round 2 NLP Model

The exact preserved Round 2 sklearn Pipeline is loaded directly: TF-IDF word 1–2 grams plus Logistic Regression. Its labels are Negative, Neutral, and Positive. The model produces predictions and class probabilities for every retained record.

SENTIMENT ANALYSIS

Predicted distribution: {'Negative': 132, 'Neutral': 38, 'Positive': 34}. Monthly sentiment index = $(\text{Positive} - \text{Negative}) / (\text{Positive} + \text{Negative} + \text{Neutral})$. Algorithmically selected descriptive candidate shifts: SS1: 2025-09→2025-10, index -0.200→-1.000 (-0.800), n=5/3; SS2: 2024-10→2024-11, index -1.000→-0.200 (+0.800), n=4/5. Small n values mean these are not statistical significance claims.

ACTIVITY ANALYSIS

Monthly record counts and available HN score totals are charted. Engagement candidate definition: the highest monthly total of available HN story scores among months with scored records and at least three retained records; reference baseline is the median total across observed scored months. ES1: 2023-03, engagement 1704 vs rolling-median baseline 6 (ratio 284.00); ES2: 2026-02, engagement 338 vs rolling-median baseline 6 (ratio 56.33); ES3: 2025-09, engagement 265 vs rolling-median baseline 6 (ratio 44.17).

TOPIC / ENTITY ANALYSIS

Topics are data-derived TF-IDF terms using English plus custom domain stopwords (min document frequency 2), not semantic topic-model labels. Highest terms: youtube, twitter, facebook, feed, algorithmic, tiktok, instagram, feeds, better, source. Entities use a documented platform/product gazetteer; X requires explicit platform context: YouTube, Twitter, Facebook, Instagram, TikTok, Google, Reddit, GitHub, Meta, Netflix.

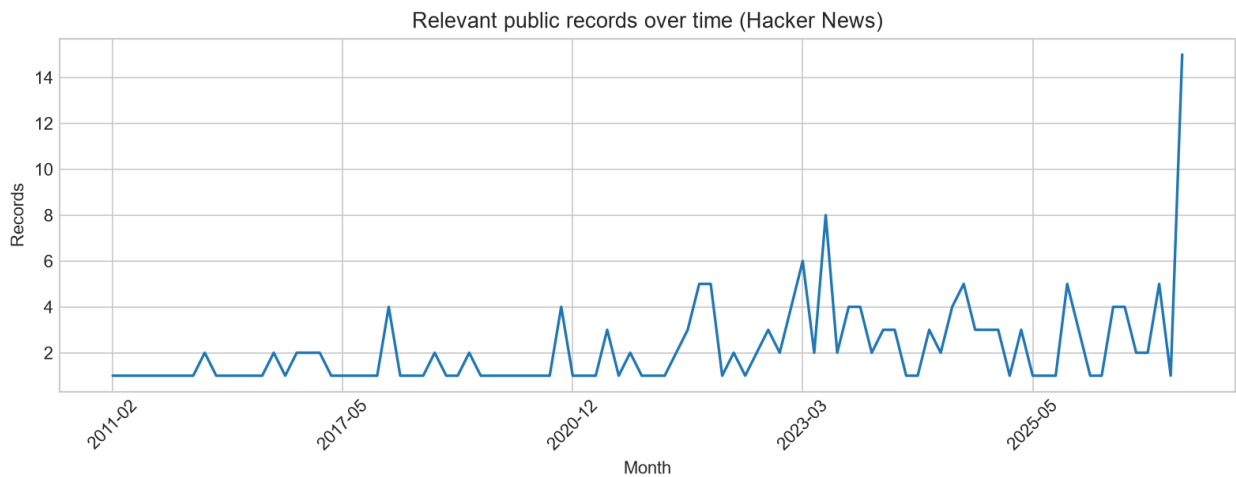
TRIGGER EXPLANATIONS

Evidence is drawn only from collected HN records: ES1: The 2023-03 engagement candidate coincided with collected HN discussion of Twitter recommendation-algorithm source code.; ES1: The 2023-03 engagement candidate coincided with collected HN discussion of Twitter recommendation-algorithm source code.; ES1: The

2023-03 engagement candidate coincided with collected HN discussion of Twitter recommendation-algorithm source code.; SS1: The 2025-09 to 2025-10 descriptive candidate shift has contemporaneous collected discussion; no specific causal platform announcement is asserted from this bounded evidence.; SS1: The 2025-09 to 2025-10 descriptive candidate shift has contemporaneous collected discussion; no specific causal platform announcement is asserted from this bounded evidence.; SS1: The 2025-09 to 2025-10 descriptive candidate shift has contemporaneous collected discussion; no specific causal platform announcement is asserted from this bounded evidence.; SS2: The 2024-10 to 2024-11 descriptive candidate shift has contemporaneous collected discussion; no specific causal platform announcement is asserted from this bounded evidence.; SS2: The 2024-10 to 2024-11 descriptive candidate shift has contemporaneous collected discussion; no specific causal platform announcement is asserted from this bounded evidence.; SS2: The 2024-10 to 2024-11 descriptive candidate shift has contemporaneous collected discussion; no specific causal platform announcement is asserted from this bounded evidence.. These observations coincide in time with the detections; they do not establish causation. All item URLs/timestamps are listed in results/trigger_evidence.csv.

Limitations, Conclusion, and Reproducibility

HN is a technically oriented public forum, rather than a representative population. This historical archive sample does not fulfill continuous live monitoring, and sparse month-level samples make descriptive shifts unstable. Re-run `python src/run_round3.py` from the project root to regenerate analysis from the stored raw export; requirements are listed in requirements.txt.



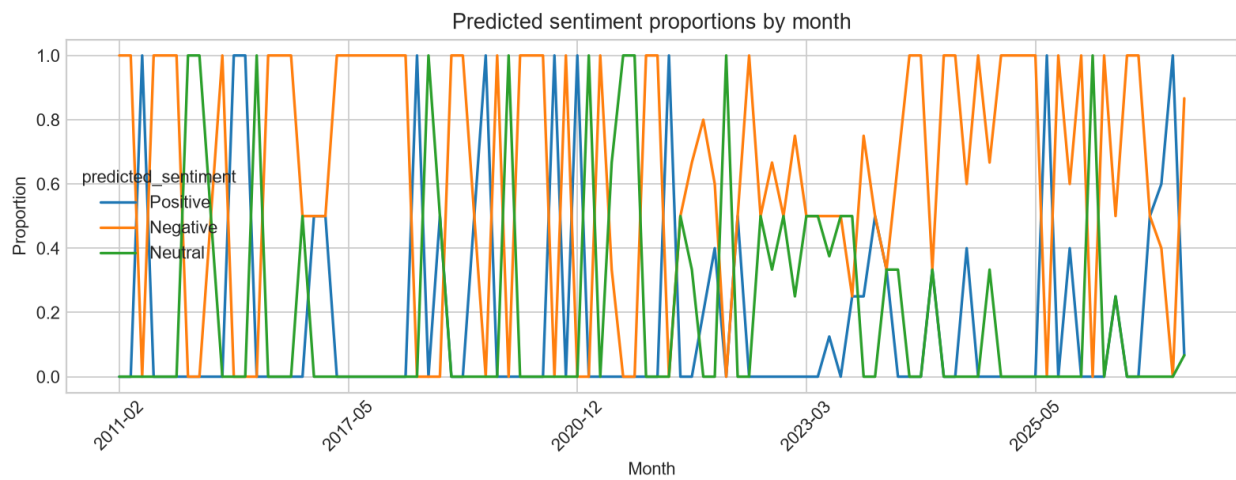


Figure 3. Monthly predicted sentiment.

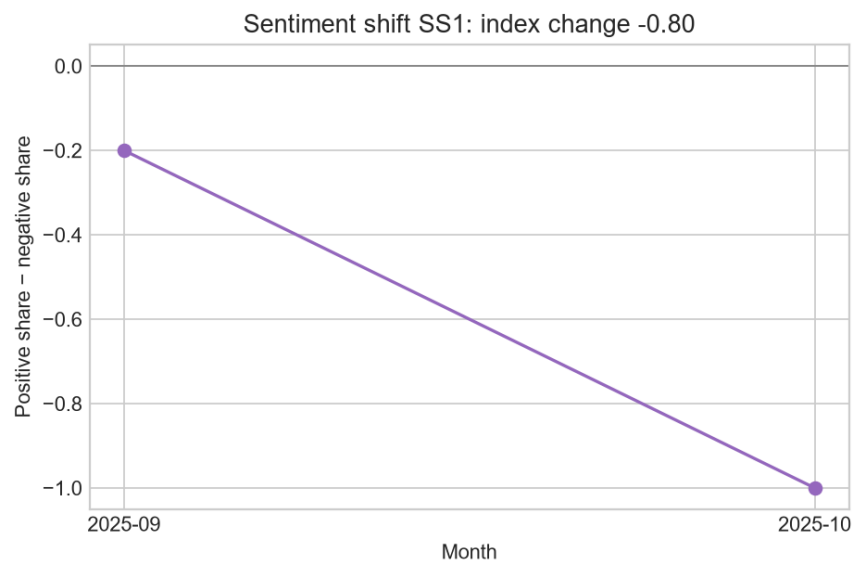


Figure 4. Descriptive sentiment shift candidate SS1.

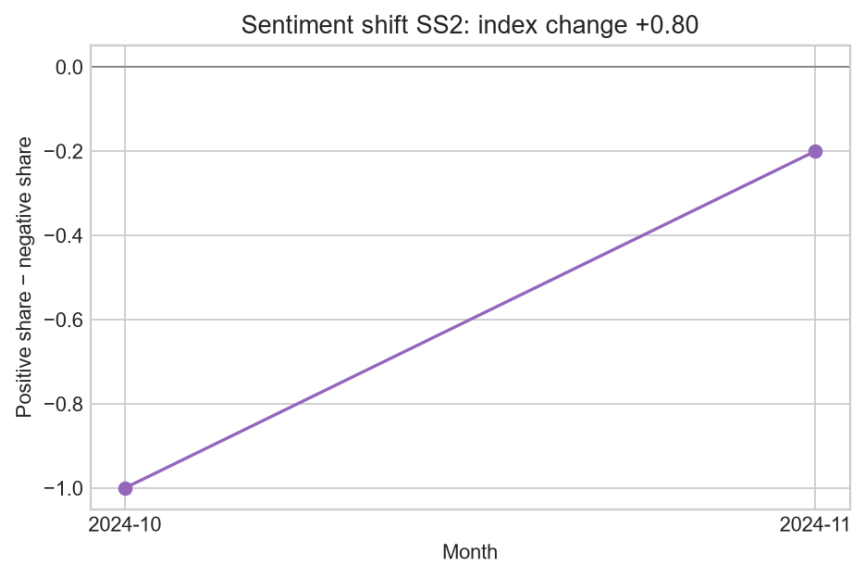


Figure 5. Descriptive sentiment shift candidate SS2.

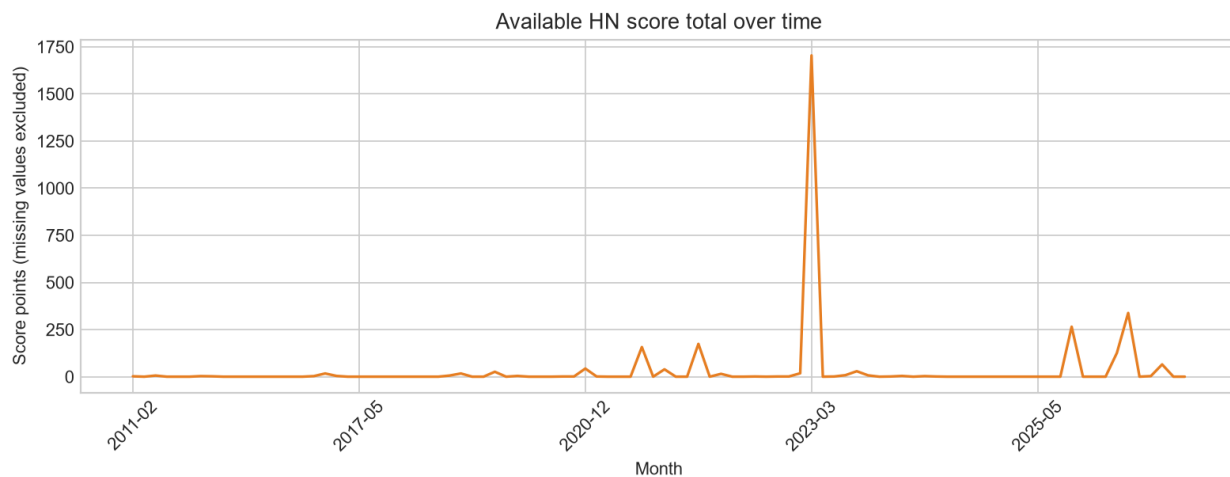


Figure 6. Available HN score over time.

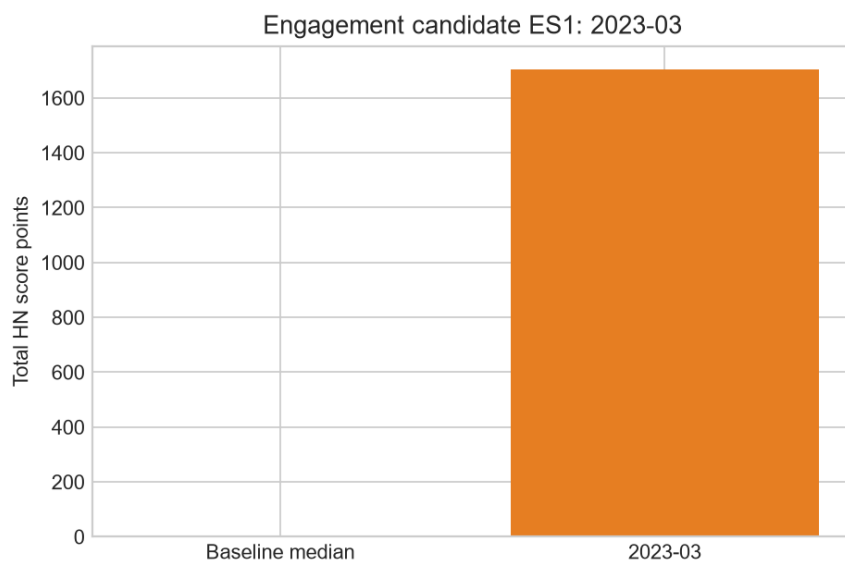


Figure 7. Engagement candidate.

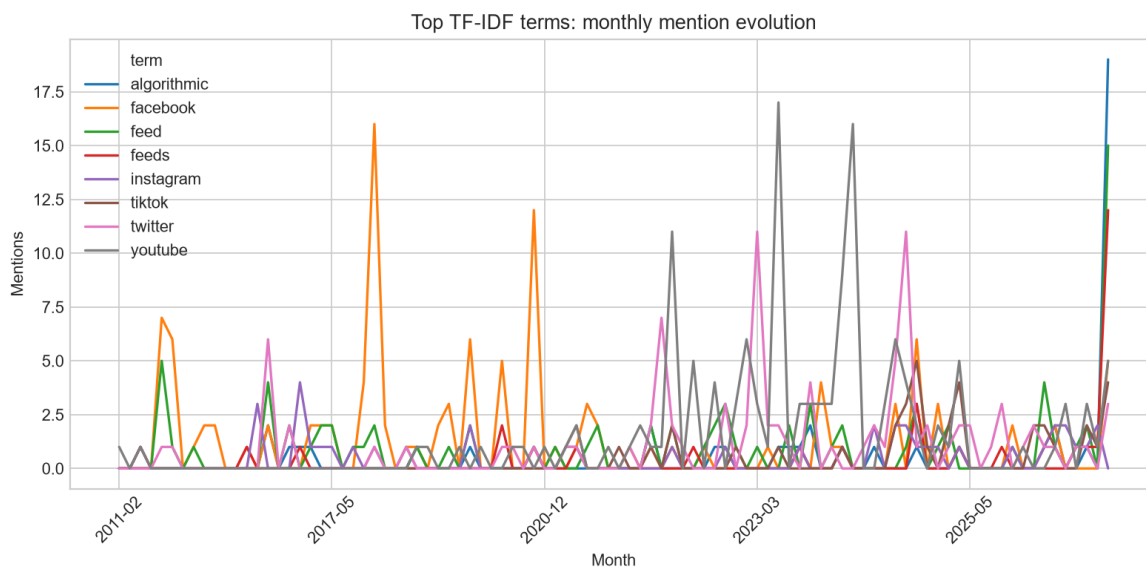


Figure 8. Data-derived term evolution.

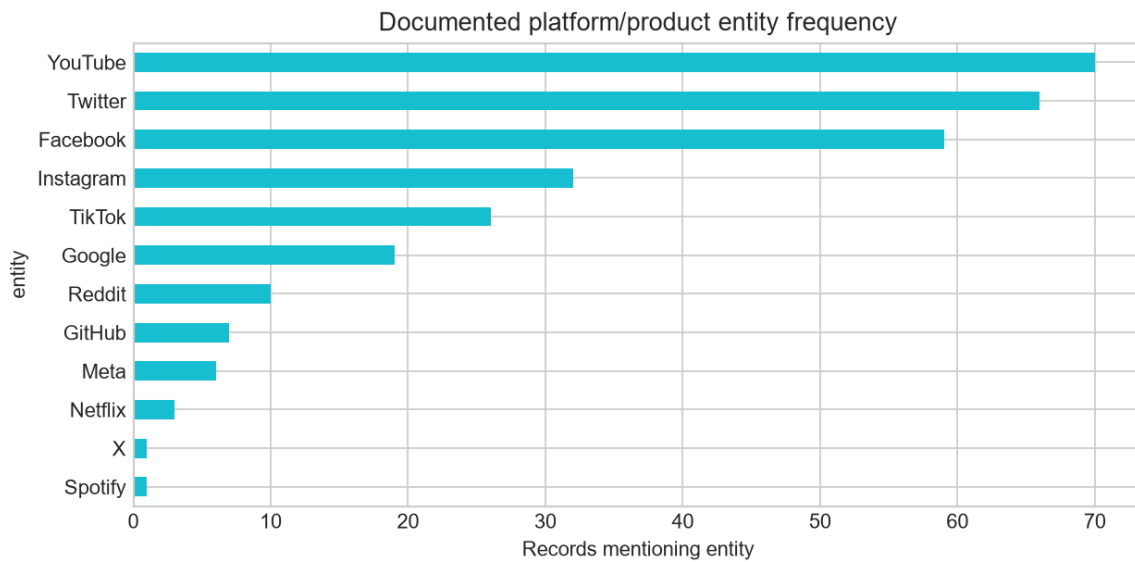


Figure 9. Entity frequency.

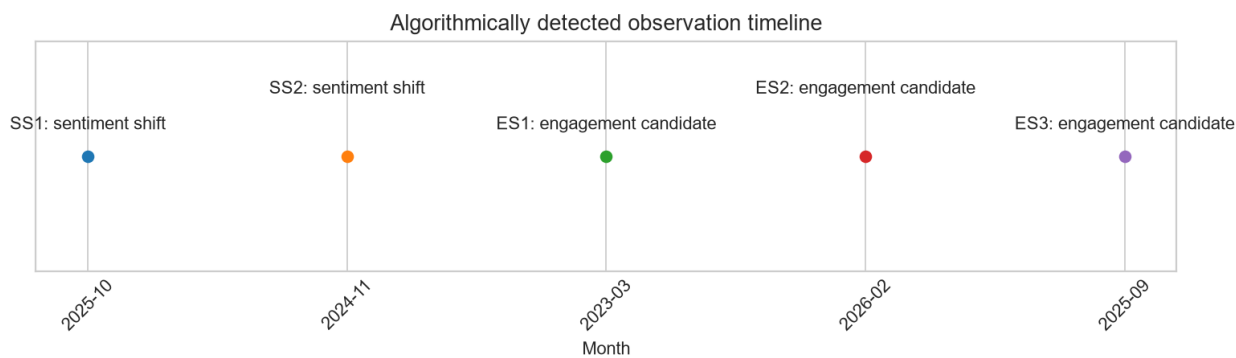


Figure 10. Detection timeline.